

VALUE ADDED COATING SOLUTION PVT. LTD.

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M/S. SKAND CREATIONS PVT LTD

Hapur (U.P.)

Kind Attention: Mr. Kapil Gupta

Subject: **Quotation for Powder Coating Plant**

Dear Sir ,

This has reference to our discussion regarding the Oven, Booth & powder coating manual Gun for the coating of Stainless Steel Utensils. We are offering you the Gas/Diesel Fired Indirect Oven , Mono Cyclone Booth with Dust Collector & Gema Manual Gun for the same purpose.

The complete details are as below as per discussion:

1.OVEN

Industrial Oven

The industrial oven plays a strategic role in many high-tech industries and in the more traditional niches of our daily lives. They cure exotic materials in the laboratory so ultra-light and super-strong materials can be processed, bake sand cores in the foundry so modern castings can be made, mass-produce delightful pastries at commercial bakeries throughout the country, preheat metals and other materials for anti-corrosion coatings and function as diffusion furnaces for the semiconductors that bring our computers to life. Our world would not be the same without the industrial oven.

Definition for Each Oven Type

An industrial oven is a heated chamber that may be used for any number of applications. **There are six basic kinds of industrial ovens:** a curing oven, a drying oven, a baking oven, a batch oven, a conveyor or continuous oven and a clean process oven. Details about each are as follows:

- **Curing ovens** are used for curing a wide variety of composite materials. They are commonly used in the aerospace and automotive industries. Curing ovens are ideal for preheating or drying parts and curing various coatings. Preheating prepares metal or other materials for the coating process.

- **Drying ovens** are essentially closed chambers designed for drying objects by heating them at relatively low temperatures. These ovens are typically used to eliminate moisture from water-based adhesives and coatings. Drying ovens are also used for removing water from the surface or interior of certain products or substances.
- **Baking ovens** can be industrial-size ovens for large food baking and pastry production. They can also be foundry core bake ovens used in iron and steel foundries to bake the sand cores used in the production of metal castings.
- **Batch ovens** come in different configurations and sizes to accommodate the needs of various industries. A batch oven can be a truck oven, a walk-in oven or a roller hearth oven. Batch ovens are typically used for processing low to moderate quantities or for applications that require long periods at a specific temperature.
- **Conveyor or continuous ovens** typically use a belt to transport foods through a cooking cavity. Conveyor ovens cook food products by blowing hot air from the top and bottom heat sources. Gas and electric versions come in countertop and floor models.
- **Clean process ovens** are typically used for die-bond curing, semiconductor packaging processing, dehydrogenation (the removal of fever-causing substances), sterilization and drying in life sciences applications. They can also be used for high-temperature applications such as the curing of polyimide films.

Industrial Heating Options

The primary types of heating for industrial ovens are **convection** and **radiation** heating.

Convection heating is carried out by heating the air in the heating chamber of the oven. This heated air is then circulated inside the oven by means of a fan or series of fans. The heat for this circulating air can come from many sources: electricity, natural gas, oil or steam. Convection ovens are suitable for both batch and continuous operation.

Normally, the temperature in a convection oven is thermostatically controlled so the heat and fuel consumption can be regulated in accordance with the work to be done. Because heat is supplied by the circulation of air, an object of any shape or size can be dried by convection heating. Convection heating is typically used for drying large casting, machined components and objects having non-uniform weight distribution.

A convection oven must be started 20 to 30 minutes before it can be used, because the air inside the oven will first have to be heated to the desired operational temperature.

Radiation heating starts with the source of heat, which is usually generated by infrared burners using natural gas or infrared bulbs and electrical heaters. This infrared radiation can be directed to the object to be heated by means of suitably shaped reflectors if necessary. The majority of the heat and infrared radiation is absorbed at the surface of the heated object. Thus, radiation heating is essentially a surface heating process ideal for curing painted surfaces.

The temperature attained by the surface of the object depends on the intensity of the radiation, the time of exposure and the mass of the article. The color of the object being heated can also play a role in how quickly the objects surface reaches the desired temperature. Black objects tend to absorb heat faster, while glossy white objects require longer exposure times.

Powder Coating Curing Oven

Powder coated substrates are ultimately indirectly cured in third and last process through hot air supplies. Powder coating curing effective metal temperature varies between 180 degree centigrade to 200 degree centigrade for time duration 10 to 15 minutes depending upon powder's technical specification and manufacturer recommendations.

In powder coating-curing oven heating media could be of mainly five kinds:

- Electrical Heaters
- L.P.G. Direct Flame Heating.
- L.P.G. Indirect R.G.T. Heating.
- **GAS/Diesel Fired Heat Exchanger.**
- Infrared Oven.

IN INDUSTRY MOST COMMONLY USED HEATING MEDIUMS ARE DIESEL/GAS FIRED HEAT EXCHANGERS (H.A.G.)

Technical Specification

Oven Dimensions(LXWXH):6'X5'X5'(Eff. inner Vol.)(Trolly Size)

Approx- Oven Outer Dimension(LXWXH) : 8' X 8' X 7'

Insulation Thickness: 200 mm

Insulation Medium: Rock Wool((96 Density)

Material Of Construction: 1.5mmCRC & 2.0mm HRC

Centrifugal Blower : 1 no. suction type

Blower Motor Rating: 5 H.P./1440 r.p.m. X 1 unit(ABB/Crompton)

Heating Media : H.A.G.(Hot air generator.)

Heat Exchanger Material : S.S. 304

Control Panel: Automatic

Burner Make : Ecoflame

2.Recovery Booth

Powder Coating Mono-Cyclonic Recovery Spray Booth

Mono Cyclonic Recovery System is a revolutionary new concept becoming popular in Indian powder coating market. Mono- cyclonic systems gives excellent 98% over sprayed powder recovery. Mono- cyclonic systems are widely used at the places where we require large volumes powder coating recovery efficiency for application.

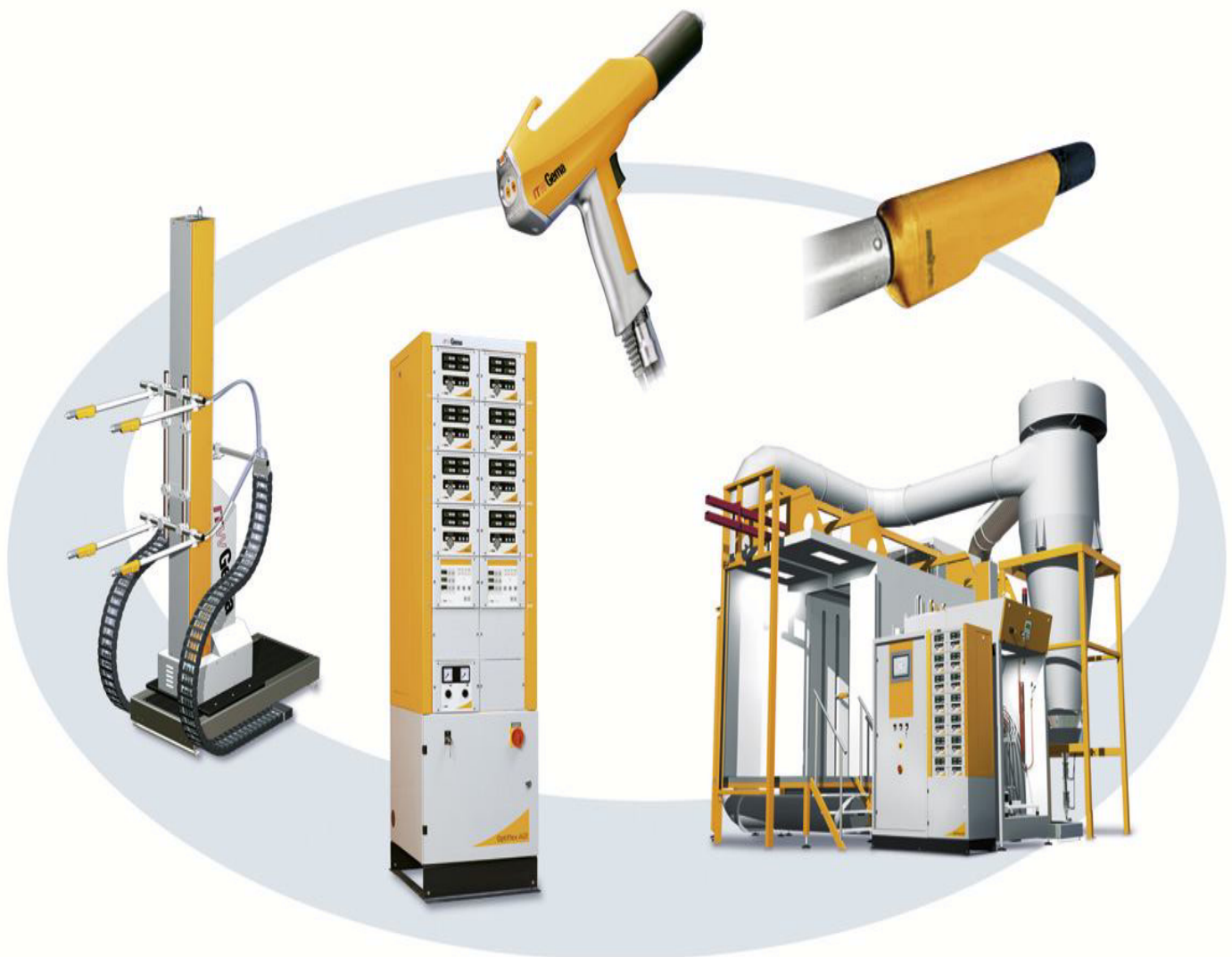
Mono Cyclonic Recovery Systems four major parts:

1. **BOOTH:** The enclosed zone in which gun operator applies virgin powder film through electrostatic charging is called booth, this is the effective volume in which powder coating process takes place.
2. **MONO-CYCLONE:** IN Mono-Cyclonic Recovery System one cyclone of large diameter introduce, depending upon the effective working volume to enhance the suction power. Since powder enters from the top of clone through the stabilizing chamber, access to clone for cleaning is extremely easy. .
3. **BLOWER:** This is mounted at the top of the dust collector to suck out the air from the booth through Cyclone along with over sprayed powder particles. Over sprayed usable powder particles get settled due to gravitational force in the collector bin at the conical end of cyclone.
4. **DUCTS:** Ducts connects booth to mono-cyclone & mono-cyclone to dust collector.
5. **DUST COLLECTOR-** Dust collector is the device in which fine powders separate from the air through the cartridge filters & collects in the bin. The motor & blower are placed in this machine. The cartridge filters clean automatically through reverse jet pulsing system.

Technical Specifications

Dimensions(LXDxH):	3' X3' X3'(Effective working volume)
Recovery System:	Mono Cyclone
Suction Motor Rating:	5 H.P. X 1unit (2800 rpm) (ABB or Crompton)
Collection of Recycled:	Cyclone
Collection of Waste Powder:	Dust Collector (For Pollution Free Booth)
Material Of Construction:	1.2mm(SS-304),1.5 mm CRC & 2.0mm HRC
Operator:	Single Operator

3.Gema Manual Gun



GEMA Manual Powder Coating Gun (OptiFlex - Pro)

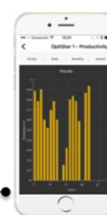


Gema

Application
Control



Line
Management



Setup



Service



A) Technical Data

A		Electrical Data	Specifications
	1	Rated input Voltage/ Connected Load	110-260 V AC/ 150 VA
	2	Frequency	50 – 60 Hz
	3	Type of Protection	IP 54
	4	Temperature Range	0 deg C to + 40 deg C
B		Pneumatical Data	
	1	Max. Input Pressure	5.5 Bar/ 80 psi
	2	Max. water vapor content of compressed air	1.3 gms/Nm3
	3	Max. oil vapor content of compressed air	0.1 mg/Nm3
	4	Max. compressed air consumption	14 Nm3/h

B) TECHNICAL DATA

1.) Application data:

Type of powder

Standard Powder

2.) Operating conditions:

Ambient air temperature range:

+10 to +40 [°C]

Work piece temperature range:

+10 to +50 [°C]

Air flow /Thermal current near the booth:

< 0.1 [m/s]

For approved coating powders only!

Technical data of the equipment can be found in the brochures.

S no.	Description	Qty	Price Each In Rs.	Total
	STAND ALONE UNIT (1 Gun, 1 Control Unit,1 Trolley & Hopper)			
1	Gema Manual Powder Coating Unit Model OptiFlex-Pro	1	1 unit	
	A. Gema OptiStar Control Unit	1		
	B. Gema Opti Gun with 6 mtrs Cable and 6 mtrs Powder Hose	1		
	C. Gema OptiFlow Injector	1		
	D. Local Powder Hopper Capacity 30 kg	1		
	E. Local Trolley	1		

COMMERCIAL OFFER

- 1. Oven- 1 Nos**
- 2. Trolleys- 2 Nos**
- 3. Booth- 1 Nos**
- 4. Cyclone- 1 Nos**
- 5. Dust Collector- 1 Nos**
- 6. Gema Manual GUN(1 Nos)**

Item 1 to 6

Rs. 14,00,000(Rupees Fourteen Lakh Only.)

Terms & conditions

Price: The above prices are ex-works Noida.

Payments: 50% against formal purchase order
and 50% prior to dispatch .

Taxes & Duties	: GST(18%) further added on to it.
Freight	: To pay basis.
Warrantee	: One Year Against any manufacturing Defect except equipment purchased by third party such as Motor,Burner etc.
Delivery	: Within 30 to 45 days from the date of formal purchase order along with advance.
Ducting :	: Ducting is not in our scope.
Civil Work	: Any Civil work if required will be at your side.

We are sure that you will find our commercial offer in line with your
acceptation and will place your valuable order with us.

For Value Added Coating Solution Pvt.Ltd.,

Harish Sharma (+919811846214)
(Director)

28 February 2025